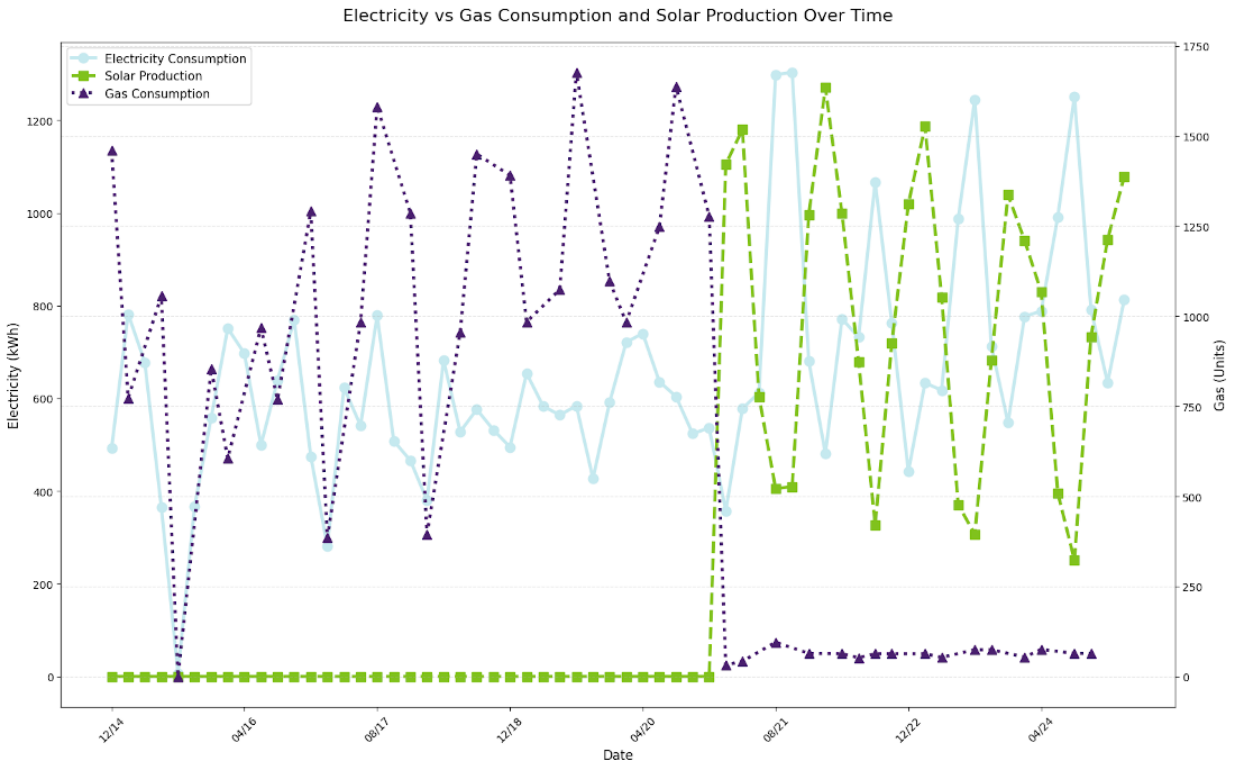


Figure 2.2 Ten years' worth of energy consumption data extracted from PDF gas and electricity bills by OCR.

Having converted all this data from PDF bills into a machine readable format, I can immediately gain insights from the data including seasonal variation on consumption, and the change in consumption when transitioning from a gas hot water system to solar hot water, as well as electricity production from solar panels installed around 2020. I will use this digitized energy data in my digital twin, but will combine it with higher resolution data from sensors moving forward.

2.3.2 Photographs

Photographs are accessible, powerful data sources, often created using just a smartphone. They provide a vital visual record and temporal sequence of changes and many organizations may already maintain extensive photographic records which can be used to reconstruct timelines of physical changes or correlate visual documentation with operational events. Photographs offer: